ENVIRONMENTAL & BIODIVERSITY PORTFOLIO

Ecology • Restoration • GIS & Biodiversity Data • Environmental Project Leadership

Selected evidence of applied ecological research, spatial analysis, project delivery and conservation impact

ARAYASELASSIE ABEBE SEMU, PhD

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Portfolio: https://magnificent-biscotti-ed1748.netlify.app/

10+ years
Postgraduate research, field ecology, teaching and environmental project leadership.

34-site Ireland programme
Research Ireland-funded forest ecology work integrating field surveys, GIS, R and remote sensing.

Field + data
Botany, vegetation surveys, habitat assessment, R, ArcGIS, QGIS, Google Earth Engine and MaxEnt.

Project leadership
Budgets, logistics, procurement, QA/QC, reporting, stakeholder engagement and team coordination.

Published evidence
Peer-reviewed outputs across forest ecology, biodiversity, climate-change modelling and ecosystem services.

Ireland-ready
Stamp 4, Irish habitat experience, own car, active CIEEM application and broad availability.

Professional Positioning

A broad environmental profile built around evidence, delivery and ecological depth.

I am a PhD-qualified environmental scientist, ecologist and project lead whose strongest value is the combination of field ecology, spatial/data analysis and disciplined project delivery. My background spans forest and wetland ecology, botany, restoration, biodiversity assessment, GIS/remote sensing, quantitative analysis, environmental reporting and stakeholder engagement. I am particularly well suited to roles where employers need someone who can move confidently from field evidence to analysis, interpretation, reporting and implementation.

Best-fit roles
Environmental Scientist; Ecologist; Biodiversity Officer; Restoration / Nature Recovery; Environmental Project Officer / Manager; GIS / Biodiversity Data; Environmental Coordinator.

Differentiator
Strong botany and vegetation ecology combined with R/GIS, project management, technical writing and senior research judgement.

Irish context
Recent UCC postdoctoral work; West Cork Fossitt habitat mapping and biodiversity planning; environmental management training.

Current development
Direct Irish statutory consultancy outputs such as NIS/EcIA/EIAR and protected-species survey breadth are developing rather than overstated.

Capability Map

What I can contribute immediately across ecology, biodiversity, data and environmental delivery.

Botany & vegetation ecology

Plant identification; floristic inventories; vegetation composition and structure; regeneration; forest and wetland ecology; habitat condition.

Habitat & restoration

Habitat assessment; restoration priorities; ecological monitoring; fragmentation and connectivity; biodiversity baselines; conservation planning.

GIS & remote sensing

ArcGIS; QGIS; Google Earth Engine; GIS in R; land-cover change; spatial data QA; GPS-referenced field data; ecological mapping.

Quantitative ecology

R; vegan; ggplot2; multivariate community analysis; diversity metrics; MaxEnt species distribution modelling; data visualisation.

Biodiversity data

Taxonomic harmonisation using World Flora Online; Darwin Core familiarity; data cleaning/validation; Git-supported reproducible research workflows.

Project delivery

Scoping; field planning; work programmes; budgets; procurement; schedules; issue tracking; QA/QC; reports; stakeholder coordination.

Communication

Peer-reviewed publications; technical reports; concise management recommendations; conference presentations; teaching; mentoring; public/community communication.

Environmental management

IEMA Foundation Certificate; ISO 14001:2015 training; environmental monitoring; continuous-improvement and evidence-management mindset.

How I Work

- Start with the decision or management question, then define the field, data and analytical evidence needed to answer it.
- Use structured QA/QC and reproducible workflows so that ecological conclusions are traceable and defensible.
- Combine field observations with GIS, remote sensing and quantitative analysis rather than treating any single evidence stream in isolation.
- Convert technical findings into practical recommendations for restoration, monitoring, planning or project delivery.
- Coordinate people, budgets, schedules and outputs with the same discipline applied to scientific methods.

Selected Professional Evidence

Government of Ireland Postdoctoral Fellowship

Led a Research Ireland-funded forest ecology programme across 34 sites at University College Cork.

Academic leadership

Former Head of Department and programme coordinator, managing staff, curricula, budgets, quality and research delivery.

Conservation project leadership

Led externally funded work on monastery forests, wetlands and habitat connectivity, including budgets and stakeholder engagement.

Applied Irish experience

Volunteer field ecologist and steering committee member supporting habitat mapping and biodiversity planning in West Cork.

Case Study 1 - Relict Forest Patch Ecology & Connectivity, Ireland

University College Cork | Research Ireland-funded programme | 34 sites | 2024-2026

Relict Forest Patch Ecology & Connectivity

Postdoctoral Researcher / Environmental Project Lead

Challenge	Small forest remnants required a robust evidence base on ecological condition, regeneration, disturbance, fragmentation and landscape connectivity to inform restoration and scientific interpretation.
My role	Led programme delivery across 34 sites, coordinating fieldwork, collaborators, schedules, budgets, data quality, analysis, reporting and research outputs.
Methods / tools	Vegetation and regeneration surveys; GPS-referenced field data; ArcGIS; QGIS; Google Earth Engine; GIS in R; R; remote sensing; fragmentation and connectivity analysis; QA/QC.
Outputs	Structured ecological datasets, spatial outputs, technical interpretation, presentations and peer-reviewed manuscripts including work on cross-boundary connectivity and wild-olive metapopulation structure.
Value	Demonstrated the ability to connect site-level ecological evidence with landscape-scale processes and translate results into restoration, monitoring and conservation priorities.

What this demonstrates

Scientific depth
Forest ecology, vegetation structure, regeneration, fragmentation, connectivity and metapopulation thinking.

Delivery discipline
Multi-site field programme management, budgets, schedules, data QA and reporting.

Spatial capability
End-to-end use of GIS, remote sensing and R for landscape ecological questions.

Employer relevance
Strong fit for restoration, forestry, biodiversity, environmental science, GIS and monitoring roles.

Case Study 2 - West Cork Habitat Mapping & Community Biodiversity

Actions for Nature Skibbereen / CECAS | 2024-Present

West Cork Habitat Mapping & Community Biodiversity

Volunteer Field Ecologist & Steering Committee Member

Challenge	Community biodiversity planning needed practical habitat information, local ecological understanding and communication that non-specialist stakeholders could use.
My role	Support field habitat identification and mapping, ecological information collation, GIS preparation, biodiversity planning and steering-group discussion.
Methods / tools	Fossitt habitat classification; plant observations; field notes; GIS mapping; spatial data organisation; community/stakeholder communication.
Outputs	Habitat information, mapped ecological evidence and practical biodiversity input for local conservation planning and habitat-enhancement discussions.
Value	Provides direct Irish-context experience in habitat terminology, field interpretation, community-facing conservation and translating ecological evidence into locally usable action.

Transferable value

- Irish habitat classification and field terminology.
- Clear ecological communication with community and non-specialist stakeholders.
- GIS-supported biodiversity planning.
- Evidence-led habitat enhancement and conservation thinking.

Case Study 3 - Northern Wollo Monastery Forests

Rufford Foundation-supported research | Vegetation ecology, ecosystem services and restoration

Northern Wollo Monastery Forests

Researcher / Project Lead

Challenge	Small culturally protected forest remnants were under fragmentation and ecological pressure but retained important biodiversity, regeneration and ecosystem-service value.
My role	Led vegetation, ecosystem-service and restoration research and integrated ecological evidence with stakeholder perspectives and conservation priorities.
Methods / tools	Plant diversity surveys; vegetation structure and regeneration analysis; community ecology; ecosystem-service assessment; stakeholder studies; R; scientific reporting.
Outputs	Peer-reviewed publications, ecological interpretation, restoration recommendations and evidence on forest-patch condition and conservation value.
Value	Shows long-term capability in relict forest ecology, restoration, stakeholder-facing research and translating field evidence into management priorities.

Case Study 4 - Babile Elephant Sanctuary

Chicago Zoological Society + Idea Wild-supported project

Habitat & Connectivity Assessment

Project Leader

Challenge	Conservation planning required stronger evidence on habitat condition, fragmentation, connectivity, restoration suitability and human-wildlife pressures.
My role	Led project coordination and habitat-focused ecological assessment, integrating vegetation and spatial evidence with field observations and stakeholder context.
Methods / tools	Vegetation assessment; GIS; landscape interpretation; habitat fragmentation; corridor thinking; restoration suitability; project and budget management.
Outputs	Conservation-planning evidence and recommendations on habitat priorities, ecological connectivity and restoration.
Value	Demonstrates externally funded project leadership, spatial ecological thinking and protected-area conservation planning.

Case Study 5 - Wetlands, Monitoring & Environmental Management

Haramaya, Teneke & Adele Wetlands | Two-year government-supported programme

Multi-site Wetland Ecology Programme

Project Leader

Challenge	Wetland management required coordinated ecological assessment, reliable data, stakeholder engagement and practical management recommendations across multiple sites.
My role	Led survey planning, field-team coordination, logistics, budgets, data quality, reporting and stakeholder engagement.
Methods / tools	Wetland ecological assessment; environmental monitoring; structured field data; ecosystem-service interpretation; stakeholder engagement; project controls.
Outputs	Technical reports, management recommendations and monitoring priorities across three wetland systems.
Value	Demonstrates end-to-end environmental project delivery beyond terrestrial forest ecology.

Case Study 6 - Protected-Area Biodiversity Baseline

Chebera Churchura National Park | Field Ecologist / Biodiversity Consultant | 2022

Biodiversity Baseline & Management Advice

Field Ecologist / Biodiversity Consultant

Challenge	Protected-area management required a baseline understanding of plant diversity, vegetation structure, ecological pressures and priority areas.
My role	Delivered biodiversity and habitat-condition baseline work and interpreted ecological hotspots, pressures and management needs.
Methods / tools	Botanical surveys; vegetation structure; habitat condition; field interpretation; biodiversity prioritisation; technical reporting.
Outputs	Baseline evidence with restoration, monitoring and management recommendations.
Value	Demonstrates applied field ecology and the ability to turn survey evidence into clear conservation-management advice.

Biodiversity Data, GIS & Quantitative Ecology

A second professional pillar alongside field ecology.

R & ecological statistics

R, vegan, ggplot2; diversity metrics; NMDS/PCA/CCA; data cleaning; visualisation; reproducible analysis.

GIS & remote sensing

ArcGIS, QGIS, Google Earth Engine, GIS in R, remote sensing, GPS, land-cover and connectivity analysis.

Species distribution modelling

MaxEnt-based habitat-suitability and climate-change analysis for endemic Vachellia negrii.

Landscape ecology

Fragmentation, connectivity, ecological networks, metapopulation structure and spatial conservation prioritisation.

Taxonomy

Routine use of authoritative taxonomic resources including World Flora Online to validate and harmonise species names.

Data standards

Familiarity with Darwin Core concepts and metadata used to structure and exchange biodiversity occurrence information.

Version control

Git used to store and manage data and analytical materials linked to recent publications and reproducible workflows.

QA/QC

Checks for missing/inconsistent values, duplicate records, taxonomic issues, spatial/coordinate inconsistencies and temporal completeness.

Illustrative Analytical Work

- Climate-change habitat suitability of endemic Vachellia negrii using species distribution modelling and spatial conservation analysis.
- Landscape connectivity and cross-boundary ecological-network analysis in fragmented forest systems.
- Plant community composition, diversity and regeneration using multivariate ecological analysis.
- Ecological and economic valuation of wetlands using field measurements and stakeholder-derived evidence.

Scientific Communication

My publication record demonstrates the full research cycle: framing ecological questions, designing studies, managing field and analytical data, testing patterns quantitatively, interpreting uncertainty and communicating results clearly. This capability transfers directly to consultancy, monitoring, biodiversity-data and evidence-synthesis roles.

Professional Experience Snapshot

Government of Ireland Postdoctoral Fellow / Postdoctoral Researcher

University College Cork | Sep 2024-Aug 2026 | 34-site forest ecology programme; fieldwork, GIS/R, QA/QC, budgets, reporting and publication.

Lecturer, Researcher, Project Lead & Former Head of Department

Haramaya University | Nov 2015-Jan 2024 | Forest, wetland, restoration and biodiversity projects; teams, budgets, teaching, programme leadership and research.

Volunteer Field Ecologist & Steering Committee Member

Actions for Nature Skibbereen / CECAS | Jan 2024-Present | Fossitt habitat mapping, GIS, biodiversity planning and community conservation.

Field Ecologist / Biodiversity Consultant

Chebera Churchura National Park | 2022 | Biodiversity baseline, habitat condition, restoration and management recommendations.

Education & Professional Development

Education

PhD Plant Biology & Biodiversity Management (2023); MSc Plant Biology & Biodiversity Management (2013); BEd Biology (2011); BA Economics.

Environmental management

IEMA Foundation Certificate in Environmental Management; ISO 14001:2015 Environmental Management Systems training.

Spatial & field development

GIS in R; Google Earth Engine; National Forest Survey / National Forest Inventory; environmental monitoring.

Professional development

Principles of Environmental Impact Assessment; CIEEM membership application submitted and currently under review.

Selected Publications

- Abebe, A., Eichhorn, M.P. & Crego, R.D. (2026). Spatiotemporal dynamics and cross-boundary connectivity of an ecological network across a fragmented landscape. Landscape Ecology.
- Semu, A.A. & Abegaz, A.W. (2026). Intraspecific genetic diversity as a driver of ecosystem functioning: trophic pathways, temporal dynamics, and conservation implications. Frontiers in Ecology and Evolution, 14, 1862503.
- Abebe, A., Bekele, T. & Lulekal, E. (2025). Plant species diversity, vegetation structure and regeneration of Northern Wollo Monastery Forests. PLOS ONE, 20(9), e0330689.
- Semu, A.A. et al. (2021). Projected impact of climate change on habitat suitability of a vulnerable endemic Vachellia negrii in Ethiopia. Sustainability, 13, 11275.

Practical Information

Work authorisation

Stamp 4 - unrestricted right to work in Ireland; no sponsorship required.

Location

Based in Cork / Skibbereen; open to suitable opportunities nationwide and hybrid arrangements.

Mobility

Own car; Irish learner permit; driving test rescheduled for late December 2026.

Availability

Available for suitable environmental, ecology, biodiversity, restoration, GIS/data and project-delivery opportunities.

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